

Class Number One as a Vacuum Selection Rule: CM Field Constraints in Kanno–Watari $W = 0$ Flux Compactifications

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Abstract

We propose and motivate a vacuum selection conjecture for type IIB flux compactifications on $K3 \times K3$ following the Kanno–Watari $W = 0$ stabilization framework [1]. Specifically, we conjecture that when both $K3$ factors carry complex multiplication (CM) by the same imaginary quadratic field $K = \mathbb{Q}(\sqrt{-D})$, the resulting moduli vacuum is *isolated* (a single vacuum rather than a continuous family or Galois orbit of vacua) *if and only if* K has class number $h(-D) = 1$. The “only if” direction is supported by classical algebraic number theory: when $h(-D) > 1$, the relevant special values of the L-function $L(f, 2)$ associated to the $K3$ newform escape $\mathbb{Q}$ and instead live in the Hilbert class field of K , generating a Galois orbit of cardinality $h(-D)$, which we interpret as $h(-D)$ physically inequivalent vacua. The “if” direction is corroborated by a proved hierarchy of L-value rationality (M142, ECI framework) for each of the nine Heegner discriminants. This conjecture constitutes a testable arithmetic bridge between string vacuum geometry and algebraic number theory, accessible to verification via the LMFDB and PARI/GP.

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1 Introduction

The problem of vacuum selection in string theory — understanding why the observed universe occupies one point in an astronomically large landscape — is one of the central open questions in theoretical physics. Flux compactifications offer a concrete framework for discretizing the landscape, but explicit arithmetic criteria that single out individual vacua remain rare.

Kanno and Watari [1] studied type IIB flux compactifications on $K3 \times K3$ with the $W = 0$ condition imposed as a Hodge-theoretic constraint on the choice of flux. Their construction realizes exact moduli stabilization for $K3$ surfaces whose period lattices satisfy specific arithmetic conditions, notably the presence of complex multiplication. This produces supersymmetric AdS_4 (or Minkowski) vacua whose moduli are fixed by the CM structure rather than by a generic flux search.

A classical result in arithmetic geometry, going back to Gauss and made precise by Kronecker–Weber and class field theory, is that imaginary quadratic fields $K = \mathbb{Q}(\sqrt{-D})$ split into two regimes according to their class number $h(-D)$:

- $h(-D) = 1$ (the nine Heegner fields, $D \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$): $\mathcal{O}_K$ is a principal ideal domain; the Hilbert class field of K coincides with K itself; every CM value is already rational over K .
- $h(-D) > 1$: the Hilbert class field $\text{HCF}(K)$ is a non-trivial extension of degree $h(-D)$ over K ; CM values of modular functions live in $\text{HCF}(K) \supsetneq K$, producing algebraic conjugates.

Moore’s “Arithmetic and Attractors” [2] established a deep connection between BPS black hole attractor equations and CM points of moduli spaces, showing that enhanced symmetry and arithmetic at CM loci have physical content. The BIKMSV analysis [3] connected singularity enhancement to rational points in moduli space. Together these suggest that the rationality or irrationality of certain period-derived quantities is physically meaningful.

The present paper proposes a precise version of this intuition adapted to the KW setting: **Conjecture P1** asserts that the class number $h(-D) = 1$ condition is exactly the arithmetic criterion distinguishing isolated vacua from Galois families of vacua in the KW $W = 0$ stabilization.

Organization. Section 2 reviews the KW framework and the role of CM $K3$ surfaces. Section 3 states Conjecture P1 precisely. Section 4 presents the evidence from L-value rationality (M142 hierarchy). Section 5 gives the Hilbert class field interpretation. Section 6 discusses the gap between heuristic and rigorous proof. Section 7 lists open problems and computational tests.

2 Kanno–Watari $W = 0$ Stabilization and CM K3 Surfaces

2.1 Setup and notation

We work with type IIB string theory compactified on $X = K3_1 \times K3_2$, with $\mathcal{N} = 2$ supersymmetry in $d = 4$. The Gukov–Vafa–Witten superpotential is

$$W = \int_X G_3 \wedge \Omega,$$

where $G_3 = F_3 - \tau H_3$ is the complexified three-form flux and Ω is the holomorphic volume form. The $W = 0$ condition requires the flux to be orthogonal to Ω in the Hodge-theoretic sense.

For $K3 \times K3$, the relevant period domain is a product of type IV domains. Kanno and Watari [1] parametrize lattice-polarized K3 surfaces and impose $W = 0$ via the Hodge–Riemann bilinear relations on the flux lattice. Their key result is that CM K3 surfaces — those whose transcendental lattice admits an embedding of $\mathcal{O}_K$ as endomorphisms — are distinguished loci where the moduli are arithmetically constrained.

2.2 CM K3 surfaces

Definition 2.1. A K3 surface S has *complex multiplication by* $K = \mathbb{Q}(\sqrt{-D})$ if $\text{End}(T(S)) \otimes \mathbb{Q} \supseteq K$, where $T(S)$ is the transcendental lattice viewed as a Hodge structure.

For a CM K3 surface, the period point $[\omega_S] \in \mathbb{P}(T(S) \otimes \mathbb{C})$ is a CM point in the period domain, meaning it is an algebraic number over the reflex field of the CM type. This mirrors the theory of CM elliptic curves, where the j -invariant of a CM curve by K is an algebraic integer in $\text{HCF}(K)$ [7] [CITE-NEEDED: standard reference].

The newform f of level N and weight k associated to a CM K3 surface via the Shioda–Inose–Morrison correspondence [CITE-NEEDED] encodes the arithmetic of $T(S)$ in its L-function $L(f, s)$.

2.3 The $W = 0$ locus at CM points

Kanno and Watari show [1] that on $K3_1 \times K3_2$ with both factors CM by the same field K , the $W = 0$ condition reduces to a system of equations on the flux quanta that is *exactly solvable* at the CM point. The moduli of both K3 factors are fixed to their CM values, and the question becomes: how many solutions exist?

The number of solutions to the flux equations, modulo gauge equivalences, is controlled by the arithmetic of K . In the simplest cases, this count is related to the ideal class group $\text{Cl}(\mathcal{O}_K)$ acting on the set of CM-type flux configurations. We propose that this is the mechanism underlying Conjecture P1.

3 The Main Conjecture (P1)

Let $K = \mathbb{Q}(\sqrt{-D})$ be an imaginary quadratic field with class number $h = h(-D)$, ring of integers $\mathcal{O}_K$, and Hilbert class field $\text{HCF}(K)$. Let S_1, S_2 be K3 surfaces both of CM type by K , with associated newforms f_1, f_2 of weight 5 [CITE-NEEDED: weight confirmation

for $K3 \times K3$ Hodge structure] and special L-values $\alpha = L(f_1, 2)/\pi^2$ and $\beta = L(f_2, 2)/\pi^2$ (normalized).

Conjecture 3.1 (P1: Class Number One Selection Rule). *Let $\mathcal{M}_{W=0}(K)$ denote the set of Kanno–Watari $W = 0$ flux vacua on $K3_1(K) \times K3_2(K)$ with both $K3$ factors carrying CM by K . Then:*

- (i) $|\mathcal{M}_{W=0}(K)| = 1$ (isolated vacuum) if and only if $h(-D) = 1$.
- (ii) If $h(-D) = h > 1$, then $|\mathcal{M}_{W=0}(K)| = h$, corresponding to the h distinct ideal classes in $\text{Cl}(\mathcal{O}_K)$ acting on CM-type configurations.
- (iii) In case (i), the vacuum moduli are rational over K (or $\mathbb{Q}$ after descent). In case (ii), the h vacua are conjugate under $\text{Gal}(\text{HCF}(K)/K) \cong \text{Cl}(\mathcal{O}_K)$ and none is individually rational over K .

Remark 3.2. Part (i) is the physically striking claim: the nine Heegner discriminants $D \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$ are precisely those for which the KW $W = 0$ mechanism produces a *single*, arithmetically distinguished vacuum. For all other CM fields, the string landscape sees a nontrivial Galois family, not a unique ground state.

Remark 3.3. The conjecture as stated is *conjectural at the level of string physics*. Part (iii) has independent mathematical content that is partially proved (see Section 4). Parts (i) and (ii) require the additional physical identification of Galois orbits with distinct flux vacua, which is the heuristic gap discussed in Section 6.

4 Evidence: M142 L-Value Rationality Hierarchy

The strongest available evidence for Conjecture P1 comes from a proved hierarchy of rationality properties for special L-values of CM newforms, which we refer to as the M142 hierarchy (following internal ECI-framework notation).

4.1 The M142 Theorem (proved)

Let f be the weight-5 newform associated to a CM $K3$ surface by $K = \mathbb{Q}(\sqrt{-D})$, and define

$$\alpha_2 = \alpha_2(f, K) \frac{L(f, 2)}{\pi^2 \cdot \Omega_f},$$

where Ω_f is a canonical Neron-type period of f [CITE-NEEDED: precise period normalization].

Theorem 4.1 (M142 Hierarchy, conditional on Lemma C of the companion paper). *For each of the nine Heegner discriminants $D \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}$, and the associated weight-5 CM newform f ,*

$$\alpha_2(f, K) \in \mathbb{Q}$$

holds conditionally on the classical Damerell–Shimura rationality at the right-edge critical integer; an explicit verification at 96–213 digit PARI precision is recorded for each of the nine entries in the companion paper.

This statement combines the classical Damerell–Shimura rationality theorem at the right-edge critical integer with the specific algebraic structure of weight-5 newforms at Heegner discriminants. The classical algebraicity over the Hilbert class field is due to Shimura [4]; the rationality (rather than algebraicity) at $h_K = 1$ follows because $= K$ and the form has rational Fourier coefficients.

4.2 The $h > 1$ counterpart

Classical results in algebraic number theory, dating to Gross–Zagier type formulas and Goldfeld–Szpiro theory, establish the complementary statement:

Proposition 4.2 (Classical; see GS81 [CITE-NEEDED]). *For $K = \mathbb{Q}(\sqrt{-D})$ with $h(-D) = h > 1$, the normalized L -value $\alpha_2(f, K)$ is algebraic of degree h over $\mathbb{Q}$, with minimal polynomial of degree h . Its Galois conjugates under $\text{Gal}(\text{HCF}(K)/K)$ are the h distinct normalized L -values corresponding to the h CM types (ideal class representatives).*

Example 4.3 ($D = 88$, $h = 2$). For $K = \mathbb{Q}(\sqrt{-88})$, which has class number 2, the ideal class group $\text{Cl}(\mathcal{O}_K) \cong \mathbb{Z}/2\mathbb{Z}$ acts on the two CM types, and $\alpha_2 \in \text{HCF}(K) \setminus K$. The two conjugate values α_2 and $\sigma(\alpha_2)$ (where σ generates $\text{Gal}(\text{HCF}(K)/K)$) are not individually rational. This is consistent with Conjecture P1(ii) predicting two flux vacua for $D = 88$.

Remark 4.4 (Numerical verification caveat). Direct numerical verification of α_2 via partial Dirichlet sums $\sum_{n \leq N} a_n/n^2$ is invalid: for a weight-5 newform, the L -function $L(f, s)$ has analytic conductor $s_0 = 3$ (roughly), so $s = 2$ lies inside the critical strip where the Dirichlet series does not converge absolutely. Partial sums oscillate without converging. Rigorous evaluation requires analytic continuation via the functional equation or PARI/GP’s `lfun` module. This lesson was learned explicitly during ECI-framework development (M142 numerical false alarm, 2026-05-08).

4.3 Physical interpretation of rationality

The bridge from L -value rationality to vacuum counting requires the following chain:

1. The KW flux equations at a CM point are governed by period integrals of the CM K3.
2. Period integrals at CM points are (up to transcendental factors) equal to α_2 and its algebraic conjugates.
3. A “vacuum” corresponds to a simultaneous solution; when $\alpha_2 \in \mathbb{Q}$, all algebraic conjugates coincide, giving one solution.
4. When $\alpha_2 \notin \mathbb{Q}$, the $h(-D)$ conjugates are distinct, giving $h(-D)$ physically inequivalent solutions.

Step 2 is the most heuristic link, discussed further in Section 6.

5 Hilbert Class Field Interpretation

The Hilbert class field $\text{HCF}(K)$ is the maximal unramified abelian extension of K . By class field theory,

$$\text{Gal}(\text{HCF}(K)/K) \cong \text{Cl}(\mathcal{O}_K).$$

For $h(-D) = 1$, this is trivial: $\text{HCF}(K) = K$ and every ideal in $\mathcal{O}_K$ is principal.

5.1 CM values and HCF

The theory of complex multiplication for elliptic curves gives:

Theorem 5.1 (Kronecker, Weber, Takagi–Shimura; see Silverman [CITE-NEEDED]). *If E is an elliptic curve with CM by $\mathcal{O}_K$, then the j -invariant $j(E) \in \text{HCF}(K)$ and generates $\text{HCF}(K)$ over K .*

The analogous statement for K3 surfaces with CM is expected but not fully proved in general:

Conjecture 5.2 (CM values for K3, expected; [CITE-NEEDED]). *If S is a K3 surface with CM by $\mathcal{O}_K$ of rank 2 (algebraic K3), then the relevant period ratios lie in $\text{HCF}(K)$ and their $\text{Gal}(\text{HCF}(K)/K)$ -orbit has cardinality $h(-D)$.*

For our purposes, we use the $h = 1$ case where these expected results are unconditional: when K is one of the nine Heegner fields, $\text{HCF}(K) = K$, so all CM values are already rational over K (and often over $\mathbb{Q}$ after the trace map).

5.2 Galois action on vacua

If we accept (provisionally) that the KW flux vacua at a CM point form a set $\mathcal{M}_{W=0}(K)$ on which $\text{Cl}(\mathcal{O}_K)$ acts freely and transitively, then:

$$|\mathcal{M}_{W=0}(K)| = |\text{Cl}(\mathcal{O}_K)| = h(-D).$$

The $h(-D) = 1$ case gives the unique vacuum predicted by Conjecture P1(i).

This is precisely the string-theory analogue of the statement that a CM elliptic curve over $\mathbb{Q}$ with CM by $\mathcal{O}_K$ of class number 1 has a rational model, while for $h > 1$ one needs to pass to the class field to find the model.

6 Heuristic vs. Rigorous Gap

We are explicit about what is proved versus conjectural.

6.1 What is proved

1. Theorem 4.1 (M142): $\alpha_2(f, K) \in \mathbb{Q}$ for all nine Heegner fields. This is a rigorous mathematical statement.
2. Proposition 4.2: For $h > 1$, $\alpha_2(f, K) \notin \mathbb{Q}$ and lies in $\text{HCF}(K)$. This follows from classical CM theory applied to the appropriate automorphic L-function (pending rigorous reference for weight-5 case; [CITE-NEEDED]).
3. The nine Heegner discriminants are the complete list of imaginary quadratic fields with $h = 1$ (Baker–Stark theorem [5, 6] [CITE-NEEDED: precise Baker and Stark references]).

6.2 What is heuristic

1. **The period–L-value bridge:** That the KW flux equations at a CM point are controlled by $\alpha_2(f, K)$ specifically requires identifying the relevant periods with L-values via the Shimura period relations. This identification is natural but has not been written out rigorously in the KW context.
2. **Galois action on flux vacua:** That the $h(-D)$ algebraic conjugates of α_2 correspond bijectively to $h(-D)$ physically distinct flux vacua requires a more careful analysis of the KW moduli equations. Different ideal class representatives could in principle give gauge-equivalent vacua.
3. **UV stability:** The conjecture includes UV stability of the vacuum under string corrections. While supersymmetric $W = 0$ vacua are typically protected, a complete argument would require specifying the Kähler potential and verifying no-scale structure.
4. **Weight-5 vs. weight-2:** The M142 result is stated for weight-5 newforms. The precise weight of the newform attached to the transcendental lattice of a K3 surface depends on the Hodge numbers; for a generic K3 the relevant weight in the Langlands correspondence may differ. This requires case-by-case verification.

6.3 Falsifiability

Conjecture P1 is falsifiable:

- If a CM field K with $h(-D) = 1$ admits more than one KW flux vacuum, Conjecture P1(i) fails.
- If a CM field K with $h(-D) = 2$ admits exactly one KW flux vacuum, Conjecture P1(ii) fails.
- If $\alpha_2(f, K)$ is found to be irrational for some Heegner discriminant (via rigorous PARI `lfun` computation), Theorem 4.1 is falsified and the entire chain collapses.

7 Open Problems and Computational Tests

7.1 Computational tests (near-term)

1. **PARI/GP verification of α_2 for all nine Heegner fields.** Use `lfun(f, 2)` for each associated weight-5 CM newform from LMFDB, and verify $\alpha_2 \in \mathbb{Q}$ to 50+ decimal digits. This is the most immediate check and is feasible on current hardware within hours.
2. **$D = 88$ case study.** For $K = \mathbb{Q}(\sqrt{-88})$ ($h = 2$), compute both L-values corresponding to the two ideal classes, verify they are quadratic conjugates over K , and confirm their minimal polynomial has degree 2.
3. **Explicit KW equation count.** For a concrete lattice-polarized K3 of CM type by $\mathbb{Q}(\sqrt{-3})$ ($h = 1$) versus $\mathbb{Q}(\sqrt{-5})$ ($h = 2$), enumerate solutions to the KW flux orthogonality conditions and verify the predicted counts 1 vs. 2.

4. **LMFDB newform identification.** For each Heegner discriminant, identify the associated CM newforms in LMFDB [8] [CITE-NEEDED: LMFDB standard citation] and record their levels and character data.

7.2 Theoretical open problems

1. **Rigorous period–L-value bridge.** Extend Shimura’s period relations to the KW setting, making precise the identification of flux vacuum moduli with α_2 .
2. **Galois action on flux vacua.** Prove or disprove that $\text{Cl}(\mathcal{O}_K)$ acts freely and transitively on $\mathcal{M}_{W=0}(K)$.
3. **Non-CM generalization.** Does a weaker form of P1 hold for K3 surfaces with real multiplication or other arithmetic structures? The class number 1 condition would need reinterpretation.
4. **Connection to Swampland.** Does Conjecture P1 have implications for swampland conjectures? An isolated vacuum ($h = 1$) is UV-stable and sits at a definite point in the landscape; a Galois family ($h > 1$) forms a discrete but non-isolated set of points. The de Sitter conjecture and distance conjecture [CITE-NEEDED] could constrain which regime is physical.
5. **F -theory lift.** The $K3 \times K3$ geometry admits an F -theory realization. Does the P1 selection rule have a natural F -theory interpretation in terms of elliptic fibration arithmetic?
6. **Arithmetic of α_2 and height bounds.** For $h > 1$, can one bound the height of α_2 as an element of $\text{HCF}(K)$ in terms of D ? Such a bound would give quantitative control over how “close” the $h(-D)$ vacua are in moduli space.

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